

3 Microtubules in the Cerebral Cortex: Role in Memory and Consciousness

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Summary. This chapter raises the question whether synaptic connections in the cerebral cortex are adequate in accounting for higher cognition, especially cognition involving multimodal processing. A recent and novel approach to brain mechanics is outlined, one that involves microtubules and microtubule-associated protein-2 (MAP2). In addition to effects on the neuronal membrane, neurotransmitters exert actions on microtubules. These neurotransmitter effects alter the MAP2 phosphorylation state and rates of microtubule polymerization and transport. It is argued that these processes are important to the physical basis of memory and consciousness. In support of this argument, MAP2 is degraded with learning in discrete cortical modules. How this relates to synaptic change related to learning is unknown. The specific proposal is advanced that learning alters microtubules in the *subsynaptic zone* lying beneath the synapse, and that this forms the physical basis of long-term memory storage because microtubule networks determine the synapse strength by directing contacts with actin filaments and transport of synaptic proteins. It is argued that this is more probable than memory-related physical storage in the synapse itself. Comparisons to consciousness are made and it is concluded that there is a link between microtubules, memory and consciousness.

3.1 Introduction

The human brain is a complex mass of tissue endowed with extraordinary capabilities. Two such phenomena are memory and consciousness. In the following chapter, I present an overview of the brain and those intrinsic molecules key to cognitive function. Microtubules turn out to be a common target of neurotransmitter action and play a significant role in learning and memory. Microtubules may also play a role in consciousness. Memory and consciousness are interrelated; thus, microtubules could be the link between these two phenomena. These are the main points covered in subsequent sections.

3.1.1 General Features of the Brain

The human brain is larger than brains of most other species, particularly animals of similar or smaller size. Recent studies using magnetic resonance imaging (MRI) verify earlier research that the typical human brain is approximately 1100 cm^3 in volume and has a surface area averaging 1700 cm^2 [67].

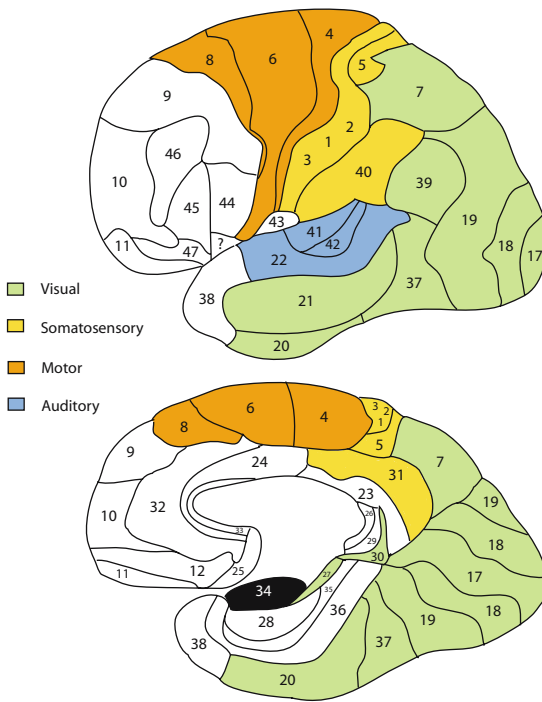


Fig. 3.1. Areas of the human brain drawn according to the scheme of Brodmann [10]. The colors refer to areas of the same (visual, somatosensory, auditory or motor) modality. Prefrontal and limbic (cingulate and parahippocampal) areas are shown in white. Olfactory area is shown in black

Similar measurements made from MRI scans show that the average cortical thickness ranges from 2.5 to 3.5 mm [29]. Given that the cerebral hemispheres are imperfect spheres with many convolutions, the cortical thickness multiplied by cortical surface area equals about half the cortical volume. Slabs of cortex are functionally divided into vertical minicolumns measuring 40–50 μm in diameter that are further linked by short-range horizontal connections into modules measuring approximately 300–600 μm^2 in diameter, and finally, modules are linked into areas by long-range horizontal connections¹. In his seminal 1909 publication, Brodmann described 47 cortical areas (Fig. 3.1).

¹ Each minimodule contains approximately 100 neurons arranged in a vertical column, an exception being that approximately 250 cells are localized to vertical columns in the primary visual area [88, 87].

The Brodmann classification system is still used today, even though some of his original cortical areas have been further subdivided.

The brain is protected in a bath of cerebral spinal fluid (CSF). Circulating CSF is especially important for the birth of new neurons and cushioning the brain. The skull confines the brain in a restricted space and the volume is kept constant. The intracranial space has a conserved mass equalizing inflow and outflow [73]. Blood flow to particular brain regions increases during specific cognitive tasks. By increasing oxygen and glucose delivery to neurons, higher levels of activity can be maintained. Methods such as positron emission tomography (PET) and functional magnetic resonance imaging (fMRI) take advantage of these changes in local energy metabolism and hemodynamics to pinpoint increases in correlated neural activity.

Two features of the human brain are believed to account for its elevated cognitive capabilities: increased number of modules in association areas and more convolutions compared to lower animals. The cerebral cortex is hierarchical in nature. Higher association areas receive projections from lower levels of sensory cortex, as well as from thalamic regions specifically supplying those regions. These higher association areas are affiliated with the most complex cognitive abilities. The prefrontal association cortex and olfactory-emotional brain circuitry occupy some of the highest tiers among cortical areas (see white areas in Fig. 3.1). Buried in the temporal lobe lies the hippocampus, a critical part of olfactory-emotional brain circuitry. Functions of the hippocampus include contextual learning², which may be similarly expressed in laboratory animals and humans. Studies on contextual fear conditioning and correlative biochemical and cellular changes in rodents will be discussed in this chapter because, for obvious reasons, it is not possible to analyze the chemical properties of cells in the human brain after learning a task.

3.1.2 Neuronal Assemblies: Patterns of Connection

The basic patterns of connection are similar across mammals: this includes rodents, nonhuman primates and humans. There is a general misconception that the mammalian cortex is completely interconnected by classical synapses³, when in fact, most cortical connections are segregated with other regions devoted to the same sensory modality (i. e. sight, hearing, touch, taste or smell). Hilgetag et al. [51] analyzed known cortical connections and found several discrete cortical clusters that are strongly interconnected. These clusters include cortical regions of the same modality; crossmodality interconnections range from sparse to absent⁴.

² Contextual learning is when an animal forms a memory of the surroundings in which an aversive or appetitive event occurs.

³ The most common synapses are between axon terminals and dendrites or cell bodies.

⁴ Some absent connections may be false negatives that could be revealed in future studies.

There are a few cortical regions where multisensory convergence has been demonstrated anatomically⁵. These multisensory areas include the posterior auditory association cortex, the superior temporal polysensory area and the intraparietal sulcus [76, 110]. It is possible that more multisensory areas will be discovered in the future.

3.1.3 Neurons, Synapses and Neurotransmitter Molecules

By definition, neurons are brain cells that are involved in communication of information. There are an estimated 100 billion neurons in the human brain at the time of birth [72]. The actual number could vary by an order of magnitude or more. Using unbiased stereology, a rigorous sampling method, 13–23 billion neurons are estimated in the human neocortex [9, 97]. Sex and age appear to be factors affecting cell number.

The neocortex has 6 layers arranged parallel to the surface of the brain, each of which contains different cell types that play different roles. Pyramidal cells populate layers III and V. Layer III pyramidal cells project to other cortical regions (corticocortical connections), whereas layer V pyramidal cells project to subcortical structures. Layer V pyramidal cells are the largest with cell diameters averaging 50 micrometer. Layer III pyramidal cells are generally smaller. Nonpyramidal local circuit cells, which do not project out of the immediate cortical area, are smaller and more numerous than pyramidal cells. These local circuit cells can be of many shapes; some are excitatory local relays and some mediate local inhibition. One type of local circuit cell, the horizontal cell, interconnects cells across the module and is generally inhibitory. Minicolumns and modules are organized around the large pyramidal cells. Sensory inputs are relayed from the thalamus to layer IV of the corresponding cortical area. The many excitatory local circuit cells relay information up and down the vertical column. In this manner, the large dendrite tree of the pyramidal cell receives many inputs via local circuit cells.

There are at least 1.25 trillion synapses in the human cerebral cortex⁶. Synapses are the tight spaces between two neurons that provide a conduit for chemical communication, but they also include the membranes on both sides of the space. Pyramidal cells receive many of their inputs onto dendrite spines, appendages that increase the surface area of dendrites and compartmentalize the molecular machinery relevant to receiving inputs. As the cortical hierarchy increases, pyramidal cells receive more inputs. As an example, pyramidal

⁵ Some demonstrated anatomical connections might be false positives due to the leakage of track tracer into neighboring regions and uptake by underlying fibers-of-passage.

⁶ Mouton et al. [89] estimated the neocortex contains only 1 billion neurons, an order of magnitude less than other reports; thus, the actual number of synapses may be higher than reported in this study.

cells in the prefrontal cortex have 23 times more spines (and inputs) than pyramidal cells in the primary visual cortex [26].

Synaptic Proteins

Synapses convey chemical messengers that alter the membrane potential of the receiving neuron. If strong enough, or if accompanied by sufficient additional input, the receiving neuron will generate an action potential that will reach another neuron. Thus, the sequence of events repeats itself. The traditional view is that the synapse is the only computationally relevant apparatus in the neuron. Hence molecules related to synaptic transmission have been studied exhaustively. Neurons generally communicate with each other using small molecule neurotransmitters released into the synapse. Many neurotransmitters are amino acids or slight modifications thereof. Major neurotransmitters in the cerebral cortex include: glutamate, GABA, acetylcholine, dopamine, norepinephrine and serotonin⁷.

Synaptic proteins relevant to receiving inputs include: (1) receptors that bind to neurotransmitters, (2) ion channels, (3) signal-transduction molecules that are activated when neurotransmitters bind to the receptor and (4) anchoring proteins that hold signal transduction molecules in place. These proteins work together to generate a local response when a neurotransmitter is released and then binds the receptor. Receptors can be ionotropic, meaning the receptor is also an ion channel. Conversely, metabotropic receptors do not have an ion channel as an integral part of the molecule. Instead, metabotropic receptors have integral G-proteins that trigger intracellular signal transduction. These intracellular signals can affect ion channels and the neuronal cytoskeleton.

Table 3.1 lists the signal transduction molecules activated by a number of metabotropic receptors.

Each cell type of the cerebral cortex uses a particular neurotransmitter for sending chemical messages. The pyramidal cells and most excitatory local circuit cells release glutamate, but these cells respond to multiple neurotransmitters. Synapses on spines usually receive glutamate inputs. Glutamate is responsible for relaying sensory information throughout the entire central nervous system. Receptors that bind glutamate are both ionotropic (AMPA, kainate and NMDA receptors) and metabotropic (mGluR). Some interneurons in the cerebral cortex use GABA, which is an inhibitory neurotransmitter. GABA_A receptors are ionotropic and GABA_B receptors are metabotropic. Acetylcholine, norepinephrine, dopamine and serotonin are considered neuromodulators because their actions are often best understood in terms of how they modify the effects of glutamate neurotransmission. Acetylcholine interacts predominantly with muscarinic receptors in

⁷ There are also a large number of neuropeptides that play specialized roles in the cerebral cortex. Due to their diversity, these will not be discussed further.

Table 3.1. Major neurotransmitters, receptors and signal transduction molecules

Neurotransmitter	Receptors	Second messengers	Protein kinases/phosphatases	Phosphorylation of MAP2
Glutamate	AMPA/kainate NMDA mGlu1, mGlu5 mGlu2, mGlu3	K ⁺ Ca ²⁺ PI-PLC ↓AC	calcineurin PKC, CaCMK II ↓PKA, presynaptic	decreased (slow) ^a increased (rapid) ^a —
Acetylcholine	M1, M3, M5 M2, M4	PI-PLC ↓AC	PKC, CaCMK II ↓PKA, presynaptic	increased ^b —
Norepinephrine	α-1 α-2 β-1	PI-PLC AC ↓AC	PKC, CaCMK II PKA ↓PKA, presynaptic	increased ^b increased ^b —
Serotonin	5-HT2 5-HT4, 5-HT7	PI-PLC AC	PKC, CaCMK II PKA	increased ^b increased ^b
Dopamine	D1, D5 D2, D3, D4	AC ↓AC	PKA ↓PKA, presynaptic	increased ^b —

^a Measured effects [104].
^b Predicted effects based on known neurotransmitter actions upon second messengers and protein kinases [108].

the cerebral cortex; however, nicotinic acetylcholine receptors are also present at low levels in the cerebral cortex. Except for nicotinic receptors, which are ionotropic, all the receptors for the neuromodulators named above are metabotropic. These receptors are linked to G-proteins that trigger an intracellular cascade of chemical responses, many of which involve the phosphorylation of key signal transduction molecules (see Table 3.1). Protein kinases responsible for phosphorylation include protein kinase C (PKC), cAMP-dependent kinase (PKA), calcium/calmodulin-dependent kinase II (CaCMK II) and mitogen-activated protein kinase (MAP kinase). Phosphatases are responsible for dephosphorylating signal-transduction molecules. One such example is calcineurin. Usually, phosphorylation activates and dephosphorylation inactivates a signal-transduction molecule; however, the phosphorylation of some proteins can interfere with polymerization of other multimolecular complexes⁸.

Neurons also contain proteins related to synthesizing, packaging and releasing neurotransmitters. Finally, there are proteins that traverse the synapse linking the presynaptic membrane with the postsynaptic membrane. Neural cell adhesion molecules (nCAMs) traverse the synapse linking the presynaptic element with the postsynaptic element. In the case of gap junctions (also known as electrical synapses) there is cytoplasmic continuity between neurons. There exists a family of connecting proteins called connexins⁹. Pannexins form another class of gap junction proteins that are found in cerebral cortex, including pyramidal cells [13].

⁸ An example is the polymerization of microtubules being decreased by the phosphorylation of microtubule-associated proteins [108].
⁹ In the early postnatal period, connexin-36 has been found in murine cortical and thalamic cells [78].

Microtubules, Microtubule-Associated Protein-2 (MAP2) and Actin Filaments

Microtubules are a class of cytoskeletal protein found in all living cells. Neurons are generously filled with microtubules, along with actin filaments and neurofilaments, two other major cytoskeletal components. Microtubules are composed of $\alpha - \beta$ tubulin dimers longitudinally arranged in protofilaments having plus and minus ends. Plus ends of microtubules undergo polymerization/depolymerization cycles to a much greater extent than do minus ends. Tubulin, the building block of microtubules, accounts for as much as 12–26% of the total soluble protein in the brain [18, 52].

Microtubule-associated proteins (MAPs) are also found in high concentrations in neurons. MAP2 is particularly abundant in dendrites where the polarities of microtubules are mixed¹⁰. Closer inspection reveals that a population consisting of many pyramidal cells and a few local circuit cells are enriched with MAP2, suggesting this cytoskeletal protein plays a prominent role in those select cortical cells involved with higher cognition¹¹. Approximately 15% of the total cortical cell population is MAP2 rich. The expression of MAP2, its phosphorylation and degradative state, varies from module to module (see Fig. 3.2)¹².

A role in cognition is further indicated by the pattern of MAP2 expression being altered by recent experience or learning; thus, the MAP2 pattern is not a permanent feature of any cortical module¹³.

The abundances of microtubules and MAP2 in dendrites, as well as MAP2 plasticity with learning, provides sound reasons to suspect a role for these molecules in information processing. For microtubules to participate in information processing, however, they must interact with synapses. The connection from synapse to microtubule is not direct. Microtubules are located in the cell body, the axon and deeper in the dendrite shaft, in a region termed the *subsynaptic zone* (the region beneath the spine neck, for spinous synapses). Typically, polymerized microtubules are absent from dendrite spines, a major postsynaptic site on neurons. The localization of actin filaments complements the localization of microtubules, however, with actin filaments filling dendrite spines. Actin filaments thus connect the synapses to the microtubules in the subsynaptic zone.

¹⁰ Plus-end directed microtubules alternate with minus-end directed microtubules in dendrites, whereas in the axon, all microtubule are aligned with the plus-end directed towards the axon terminal.

¹¹ MAP1, MAP5, Neurofilament proteins and muscarinic acetylcholine receptor also enrich in this cortical population [147].

¹² MAP2 is in a state of continuous flux with respect to its phosphorylation and degradation.

¹³ Different kinds of training alter MAP2 immunostaining in cortical modules [144, 145, 130].

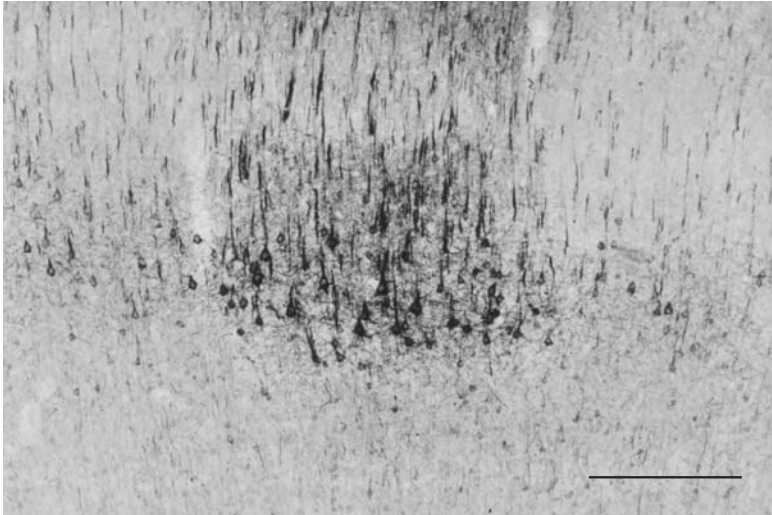


Fig. 3.2. Module of neocortex in rat brain having elevated immunohistochemical reaction for MAP2. Mainly the large layer V pyramidal cells and their thick apical dendrites extending towards the surface of the brain are intensely stained, especially in the middle module. This intensely stained module is flanked by other cortical regions that are less intensely stained for MAP2. Scale bar equals 200 μm . Adapted from [148].

MAP2 and MAP1B have been shown to provide a link between microtubules and actin filaments [21]. This bridging mechanism is a dynamic one. MAP2 alternately binds microtubules or actin filaments, often sequentially. Also, it is a dynamic pool of actin filaments that attaches to the postsynaptic density [45]. For MAP2 to bind microtubules efficiently, it must be partially dephosphorylated. Protein kinases regulate phosphorylation and the binding process. PKA, for example, favors MAP2 dissociation from microtubules and association with actin [96]. In this manner, signal-transduction molecules regulate the switching of MAP2 from microtubules to actin filaments.

3.2 Functions of Microtubules and MAPs

Microtubules were first described as providing structure to the cell. This basic function can be related to information processing, insofar as modifications to dendrite structure and branch geometry will alter the topography of inputs and thereby influence neuronal computation. Today, however, we know much more about the biochemistry and biophysics of microtubules. Our increased understanding of the molecular underpinnings of basic microtubule function is beginning to elucidate precisely how microtubules can participate in information processing.

3.2.1 Transport along Microtubules

Transport occurs along microtubules in dendrites and axons in both anterograde and retrograde directions via distinct motor proteins. Transported cargos include receptors, signal-transduction molecules, cytoskeletal proteins and mRNA. Brain-specific motor proteins include examples from the superfamily of kinesin proteins (KIFs), each of which binds to a specific cargo determined by specific amino acid sequences in the tail region and usually transports cargo towards the plus end of the microtubule [90]. Dynein is a microtubule motor that transports neurofilaments to the minus ends of microtubules [112]. Myosin-V, yet another motor protein, may facilitate the transfer of cargo from microtubules to actin filaments [12].

Kinesin-mediated transport along microtubules is affected by the biophysical state of the microtubule, as determined by posttranslational modifications and binding to MAP2 and neurofilaments. Kinesin binding to microtubules is enhanced by polyglutamylation of tubulin when up to three glutamate residues are added [137]. Adding more than three glutamate residues interferes with kinesin binding. Kinesin binding to microtubules is hindered by MAP2 also being bound due to steric interferences caused by both molecules binding near the C-termini of tubulin [2]. Once bound to the microtubule, kinesin processivity along a microtubule protofilament is unaffected by MAP2, the interpretation being that MAP2 lays down as kinesin walks over it [111]. Individual strands of MAP2, not reinforced by pairing with another MAP2, are flexible enough to be capable of lying down on the microtubule as suggested. Highly stabilized microtubules, for example, those connected to phosphorylated neurofilaments, have decreased transport rates among kinesins¹⁴. Kinesin also exerts biophysical effects on the microtubule. This motor contributes to the instability of microtubules as it interferes with tubulin polymerization [56]. Although the situation is far from being simple, an inverse relationship often exists between transport and microtubule stability.

3.2.2 Signal Transduction and Anchoring of Signal-Transduction Molecules

In addition to being a structural protein, MAP2 has been identified as a signal-transduction molecule. MAP2 can be activated through phosphorylation and inactivated through dephosphorylation. MAP2 is also an anchoring protein, meaning that it tethers, or fastens, other signal-transduction molecules together. MAP2 has also been identified as a gelation factor, rigidifying actin filaments and microtubules into a gel [68, 40].

There are over 40 sites that can be phosphorylated or dephosphorylated on MAP2. MAP2 is phosphorylated by multiple protein kinases: being an excellent substrate for CaCMK II, PKC, PKA and MAP kinase, among others [108]. CaCMK II mediates phosphorylation at as many as 18 binding

¹⁴ This is the case in axons (see Shea [114]) and remains to be studied in dendrites.

sites on the MAP2 protein, and PKC mediates phosphorylation at as many as 15 sites [133]. PKC is responsible for phosphorylation at binding sites Ser-1703, Ser-1711 and Ser-1728, which dramatically inhibits MAP2 binding with microtubules [1]. The N-terminus of MAP2 is an anchoring site for the regulatory subunit II of PKA [48]. In this manner, MAP2 is able to trap and disable PKA, and then release and enable PKA when it is necessary to phosphorylate microtubule proteins. The phosphorylation state of MAP2 regulates the polymerization and depolymerization of microtubules [108]. Phosphorylated MAP2 is much less efficient at stimulating the polymerization of microtubules; however, even in the absence of MAP2, tubulin can polymerize into microtubules at a slow rate. Calcineurin dephosphorylates MAP2.

In the brain, these signal-transduction pathways, of which MAP2 is an integral part, are triggered by neurotransmitter actions arising from synaptic inputs. These neurotransmitter inputs utilize a number of receptors triggering specific signal-transduction cascades (see Table 3.1). These actions will be elaborated on in the next section.

Glutamate Effects on Microtubules Mediated by AMPA, Kainate, mGlu and NMDA Receptors

Glutamate mediates sensory input to the cerebral cortex, where many of these excitatory presynaptic terminals contact spines. AMPA and kainate glutamate receptors respond quickly, within a few ms, altering ion conductances across the membrane (Na^+ influx $\gg$ Ca^{2+} influx). Under physiological conditions, there are a few ways that AMPA receptors (and similarly kainate receptors) can influence microtubules (see Fig. 3.3).

AMPA receptor activation can affect microtubules directly through ionic currents that passively diffuse along the membrane and penetrate the subsynaptic zone. The outermost microtubules are roughly a hundred nanometers below the membrane surface, such that these microtubules will be most readily affected by ion influxes. Possible consequences of AMPA receptor activation (and subsequent increased Na^+ influx) are initial increases, and eventual decreases, in microtubule polymerization expected to occur as the Na^+ concentration in the cell adds to the already high K^+ concentration inside the neuron (see Fig. 3.3).

Sodium or potassium ion concentrations of 150–160 mM increase microtubule polymerization, whereas Na^+ or K^+ concentrations over 250 mM decrease microtubule polymerization [92, 141]. Thus, ionic strength exerts tight bidirectional control over microtubule polymerization and thereby stands to affect other biophysical aspects of microtubules. MAP2, which crosslinks microtubules and actin, is also sensitive to pH shifts¹⁵.

¹⁵ An ATP and Mg-dependent, serine/threonine-specific MAP2 kinase can be converted to a phosphatase with certain pH changes [150].

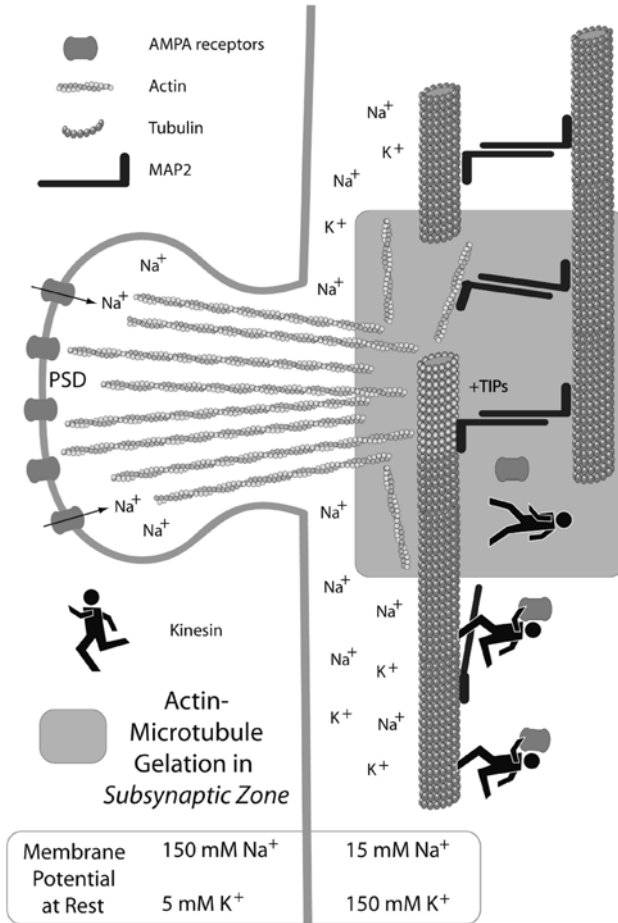


Fig. 3.3. AMPA receptor activation: effects on microtubules. A population of AMPA receptors located on a dendrite spine is shown. Microtubules are diagrammed as orange polymers composed of tubulin. Pairs of MAP2 molecules link microtubules, whereas single MAP2 molecules are flexible enough to allow kinesin to walk over. The following responses are illustrated: AMPA receptor activation opens sodium (Na⁺) channels and sodium ion influx increases the sodium ion concentration inside the neuron, adding to the potassium ion (K⁺) concentration (insert lists sodium and potassium concentrations at rest). This is likely to initially increase polymerization (indicated by green tubulins) of the nearest microtubules located approximately 100 nm beneath the spine or neuronal membrane. Plus-end binding proteins (+TIPs) will stabilize newly polymerized microtubule ends. If the sodium concentration becomes too high, however, microtubule depolymerization will occur. Actin filaments connect the postsynaptic density (PSD) to microtubules. AMPA receptor activation is likely to increase actin-filament interaction with microtubules (actin-microtubule gelation) in the subsynaptic zone, and this is likely to slow transport along microtubules. Decreased transport near the spine may facilitate unloading cargo, conceivably providing more AMPA receptor to the spine

The second way in which activating AMPA receptors could affect microtubules is indirectly, through actin filaments. Large spines that are enriched with AMPA receptors contain many actin filaments. Multidomain scaffolding proteins link actin filaments with the postsynaptic density [103], and actin links to microtubules. The interaction between microtubules and actin is likely to involve many proteins, including MAPs and motor proteins¹⁶. A functional relationship between AMPA receptors and MAP2 exists. AMPA administered at cytotoxic levels exerts long-term effects, maximally increasing MAP2 at 5 days [71].

The third way that AMPA receptor activation might influence microtubules is through actin filament–microtubule gelation. Actin filament–microtubule gelation is mediated by MAP2, among other molecules [40]. When MAP2 complexes to actin filaments and microtubules, the viscosity of the complex increases by one hundred-fold or more. Glutamate, acting through ionotropic receptors, inhibits actin dynamics and decreases filopodia motility in dendrites and axons [19], an effect that appears to reflect increased viscosity. Increased polymerization of microtubules in the subsynaptic zone would also increase interaction between actin filaments and microtubules and thereby further favor the formation of highly viscous MAP2-actin–MAP2-microtubule complexes (see Fig. 3.3).

Increased viscosity is known to slow the transport of kinesin [33]. An AMPA-enriched spine cannot operate at full capacity without replenishing AMPA receptors. The transport of AMPA receptors depends on both actin motors and microtubule motors,¹⁷ indicating that coordination between these cytoskeletal components is necessary for proper function. Increased viscosity would impede this process. To summarize, the net result of AMPA (and kainate) receptor activation is likely to increase polymerization of microtubules, perturb actin dynamics and slow transport along microtubules. Decreased transport near the spine may facilitate unloading cargo, such as the AMPA receptor (see Fig. 3.3).

Upon release of glutamate at the spinous synapse, mGluRs are also activated. At least eight mGlu subtypes have been identified, many with further subdivisions within subtypes. A certain class of mGlu receptors mediates rapid phosphorylation of MAP2 via CaMK II and PKC [104]. These kinases are triggered by the PI-PLC signal-transduction pathway and mGlu1 and mGlu5 receptors. As shown in Fig. 3.4, mGlu activation would be expected to phosphorylate MAP2, which in turn would interfere with the polymerization of microtubules in the subsynaptic zone. Consistent with this, mGlu1 α has been shown to inhibit microtubule formation by a mechanism that depends on the PI-PLC pathway [57]. Since metabotropic receptors can

¹⁶ Microtubule–actin interactions studied during neurite formation implicate MAPs [21].

¹⁷ Kim and Lisman [66] were able to interfere with AMPA receptor transport by inhibiting dynein, an actin motor, and kinesin, a microtubule motor.

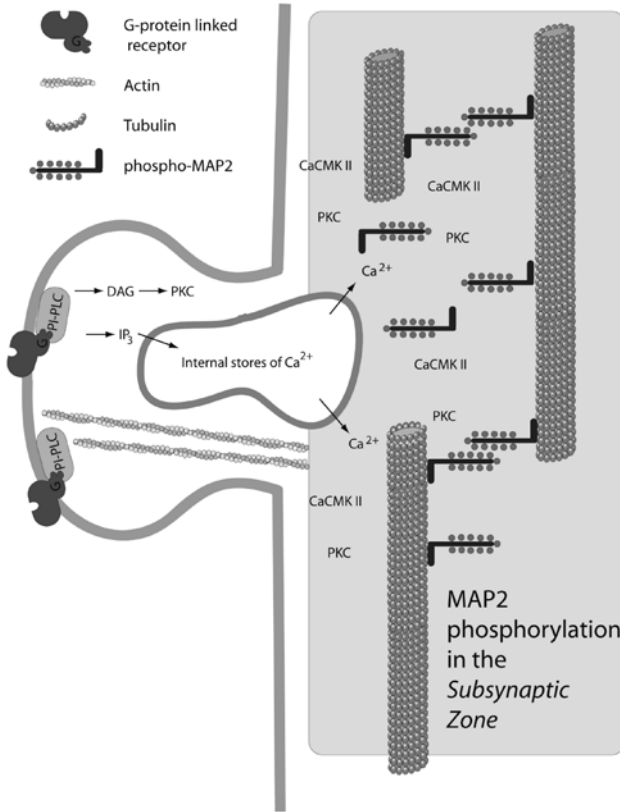


Fig. 3.4. Metabotropic glutamate (mGlu) receptors activate PI-PLC; this in turn activates diacylglycerol (DAG) and inositol triphosphate (IP₃), which in turn activate PKC and release endogenous Ca²⁺. Upon its release, endogenous Ca²⁺ activates CaMK II. Both PKC and CaMK II phosphorylate MAP2, and this interferes with microtubule polymerization in the subsynaptic zone beneath the dendrite spine. Microtubules repel one another when MAP2 is highly phosphorylated

activate many second messengers, widespread inhibition of microtubule polymerization would be expected.

Released glutamate will also bind NMDA receptors. The NMDA receptor acts as a calcium channel and its activation results primarily in the influx of calcium (see Fig. 3.5). Calcium ionic fluxes caused by NMDA receptor activation have penetrating effects that reach the mitochondria, which can be located deep beneath the membrane [134], so NMDA effects on microtubules via increased calcium ion concentration are likely. The activation of the NMDA receptor stimulates a number of calcium-activated proteins. These include: calmodulin (a ubiquitous Ca²⁺-sensing protein), calpain (a protease), CaMK II and calcineurin. NMDA receptor activation causes a persistent

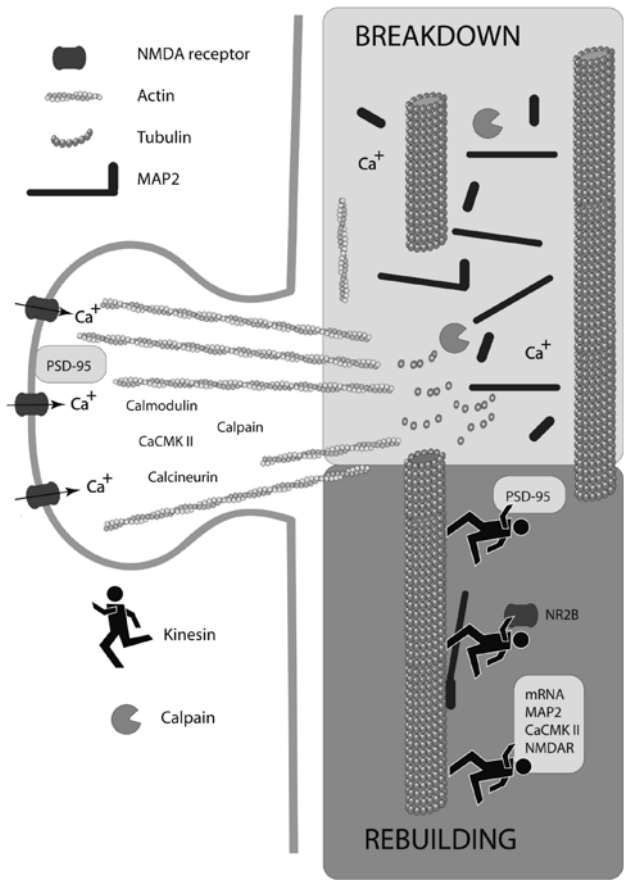


Fig. 3.5. NMDA receptors act as calcium (Ca^{2+}) channels. Calcium enters the neuron and activates calmodulin, calcium/calmodulin kinase II (CaMK II), calcineurin and calpain. Calpain is responsible for the proteolytic breakdown of MAP2 and may also lead to the breakdown of tubulin. Calcineurin causes a long-lasting increase in MAP2 dephosphorylation (not illustrated), which may enhance microtubule polymerization during rebuilding. Other events occurring during rebuilding include increased transport of postsynaptic density protein, PSD-95, and NMDA receptor subunits, e.g., NMDA receptor subunit 2B (NR2B). Rebuilding also includes increased transport of mRNA for MAP2, CaMK II and NMDA receptor (NMDAR), essential elements of the overall complex

dephosphorylation of MAP2 mediated through calcineurin [104]. Calcineurin activation by NMDA receptors would favor enhanced polymerization of microtubules. However, the proteolytic effects of calpain degrading both MAP2 and tubulin counterbalance this effect. Thus, NMDA receptors appear capable of exerting opposing effects on microtubule polymerization.

Additional evidence supports NMDA receptors mediating breakdown and then rebuilding of glutamate synaptic sites on spines. Hence, this receptor is key to neuroplasticity. Fundamental to this role, NMDA receptors form complexes with the postsynaptic density protein, PSD-95 [85], with F-actin via the actinbinding protein, α -actinin [149], and with MAP2 [14]. Activation of NMDA receptor is critical in triggering the transport of MAP2 mRNA to dendrites [58]. Kinesin-mediated transport along microtubules occurs for the NMDA receptor subunit NR2B [42] and for PSD-95 [85]. As illustrated in Fig. 3.5, direct and indirect interactions between NMDA receptors and the cytoskeleton are key in the breakdown and rebuilding of the subsynaptic zone.

Acetylcholine, Dopamine, Norepinephrine and Serotonin Trigger G-Protein Cascades that Phosphorylate MAPs

The neuromodulators – acetylcholine, dopamine, norepinephrine and serotonin – are often released onto dendrite shafts rather than onto spines. Their actions are slower than the actions of glutamate, but by no means less influential. Glutamate provokes a large membrane potential change, but this decays as it passively diffuses along the dendrite. Acetylcholine, on the other hand, produces modest membrane potentials, but these are nondecremental, indicating that a self-propagating intradendritic mechanism is involved¹⁸. This intradendritic mechanism relies, in part, upon G-protein receptor signal-transduction cascades. Because of G-protein cascades, acetylcholine and monoamines could, in principle, alter MAP2 phosphorylation all along the dendrite, thereby affecting widespread microtubule stability. Not all parts of the microtubules are equally unstable, however. Stable linkages at minus ends of microtubules are less susceptible to MAP2 binding alterations influencing microtubule stability (see Fig. 3.6).

Unstable MAP2 linkages, however, would be affected as far as signal-transduction molecules and protein kinases were able to passively diffuse. Since the concentration of these would decline with distance, other mechanisms may be involved. To the extent that MAP2-tubulin binding affects the biophysical state of tubulin¹⁹, that biophysical state of tubulin could, in principle, be propagated to other tubulins down the microtubule. Deformations in dimer–dimer interactions are suggested, which begin at minus ends, travel the length of the microtubule and promote destabilization or depolymerization at the plus ends [109]. These kinds of biophysical effects could conceivably account for the fact that acetylcholine produces effects that extend a long distance along the dendrite shaft.

¹⁸ Acetylcholine iontophoresed onto pyramidal cell dendrites produced effects that remained equal in amplitude along the entire dendrite, in contrast to effects of glutamate, which decayed dramatically at increased distance from the infusion site [83].

¹⁹ Biophysical states of tubulin include parameters such as dipole moments, conformational states, and surface electrostatic charge.

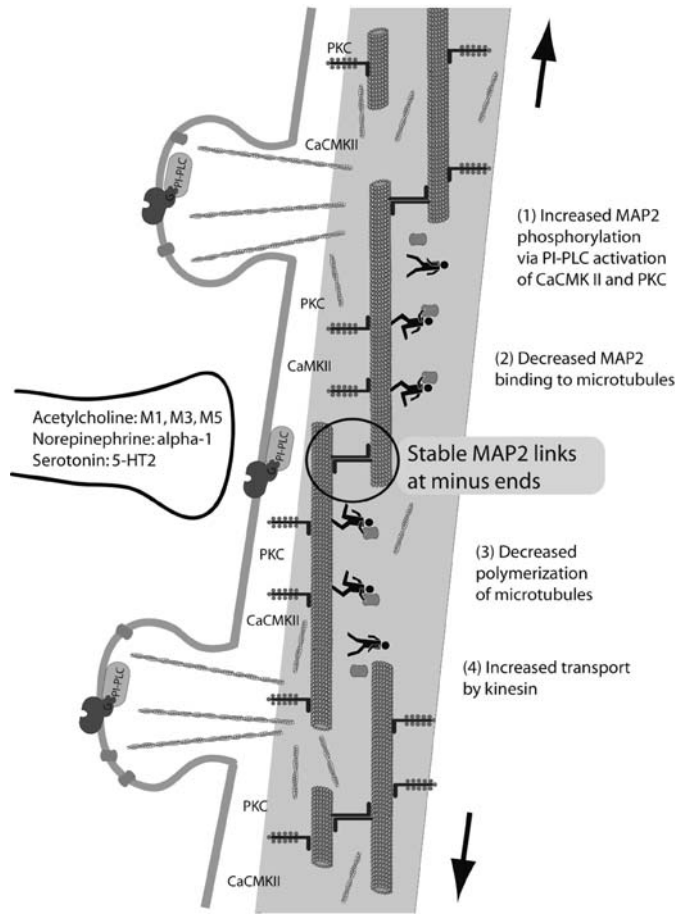


Fig. 3.6. Terminals releasing acetylcholine, norepinephrine and serotonin generally contact the dendrite shaft. Illustrated receptor actions include acetylcholine-M1, M3 and M5; α 1-adrenergic and serotonergic-5-HT2, all of which activate PI-PLC, which in turn activates CaMK II and PKC. The subsynaptic zone that is affected is much more extensive for these nonspinous inputs; these action may extend the entire length of the dendrite. Phosphorylation of MAP2 will decrease polymerization of microtubules and cause microtubules to repel one another. This would be expected to increase the rate of transport by kinesin. Periodic MAP2 links would be expected to remain intact, and the highly stable MAP2 links between minus ends of microtubules may remain intact even near the release sites of acetylcholine, norepinephrine or serotonin. See Figs. 3.3–3.5 for definition of symbols

Acetylcholine muscarinic receptors are colocalized within the same neurons as MAP2 and PKC, and these proteins appear to form multimolecular complexes [129, 147]. Acetylcholine muscarinic receptors M1, M3 and M5, serotonin 5-HT2 receptors, and α 1-adrenergic receptors activate the PI-

PLC pathway (similarly to mGlu1 and mGlu5). The PI-PLC pathway then activates diacylglycerol (DAG) and inositol triphosphate (IP3), which, respectively, activate PKC and release endogenous Ca^{2+} . Upon its release, endogenous Ca^{2+} activates CaMK II. Both PKC and CaMK II phosphorylate MAP2 [108]. Thus, the PI-PLC pathway, which is strongly activated by acetylcholine (and to a lesser extent by serotonin and norepinephrine), is capable of exerting powerful effects on microtubule polymerization via phosphorylative effects on MAP2. Alterations of microtubule polymerization will influence microtubule transport. These molecular dynamics effects may underlie the strong involvement of acetylcholine in memory, attention and dream sleep.

A number of additional receptors activate PKA. These include: dopamine D1 /D5, β 1-adrenergic, and serotonin 5-HT3, 5-HT4, 5-HT6 and 5-HT7 receptors, all of which increase levels of cAMP, and this in turn activates PKA. PKA is known to phosphorylate MAP2 [108], and PKA also stimulates the binding of MAP2 to actin and its dissociation from microtubules [96]. MAP2 alternating between the binding to microtubules and the binding to actin filaments might be important for keeping the microtubule network linked with synaptic activity. This would be consistent with the role of norepinephrine (and other monoamines) in vigilance, a heightened state of awareness of environmental stimuli. Receptors that decrease cAMP, due to the fact that they are negatively coupled to adenylyl cyclase, are frequently located on presynaptic terminals, in which case they do not come into contact with dendritic microtubules or MAP2. Thus, one would only expect these receptors to be involved in dendritic information processing to the extent that they are present on dendrites or to the extent that they regulate presynaptic neurotransmitter release and this release, in turn, has effects on the postsynaptic cell.

Those multiple presynaptic sites releasing acetylcholine or monoamines emerge from single axonal fibers often following the entire dendrite length, which enables these inputs to modulate long stretches of microtubules. This anatomical match to the intradendritic network of microtubules may make these synapses anatomically well suited to convey the global states involved with learning, memory, attention and consciousness. These global states will be discussed in the subsequent sections.

3.3 Learning and Memory: Neuroplasticity vs. Stability

The brain changes whenever an animal learns an association or when an operant behavior is strengthened because of a reinforcer²⁰. More than mere accompaniment, it is assumed that the change in the brain is responsible for the change in behavior. Many kinds of neuroplasticity are found during early

²⁰ During operant conditioning, positive reinforcers increase “reward-producing” behaviors and negative reinforcers increase behaviors such as escape and avoidance.

development and with learning that could account for the behavioral changes observed. The expectation of neural plasticity is fulfilled by supportive experimental data.

Once a change has occurred, however, then it becomes important to stabilize the neural substrate that is responsible for the recently acquired, adaptive behavior. Thus, the same neuronal compartments must be involved in both change and stability or the two compartments must interact. Experimental data do not consistently indicate that changes in synapses are stable. Thus, a new paradigm may be needed to replace the long-held notion of information storage in synapses. The new learning and memory model advanced here is that information storage occurs in the subsynaptic zone, wherein microtubules determine synaptic properties based on previous synaptic activity.

3.3.1 Synaptic Change: Hebb's Rule Revisited

Since Donald Hebb [50] first proposed that synaptic knobs might reorganize with learning, there have been hundreds of studies examining brain tissue for synaptic reorganization following learning or following comparable physiological stimulation that might resemble learning-related events²¹. I will preferentially address those studies that assessed synapse and spine changes accompanying learning in behaving animals.

Temporary Nature of Synaptic Change

Researchers have studied synaptic change with learning using either the electron microscope to visualize synapses or light microscopy to visualize spines. Numerous studies have found evidence for change in spine morphology with learning or enriched experiences (for a review, see Yuste and Bonhoeffer [151]). Regarding the more specific issue of synaptogenesis²², recent studies have shown that synapse and spine densities are altered following learning. In a recent study of associative learning, adult male rats trained on a trace eye-blink conditioning task showed increased the density of spines on basilar dendrites of pyramidal neurons in CA1 of hippocampus [75]. Apical and basilar dendrite lengths were not altered by the training, indicating that the increased spine densities represented increases in total numbers of spines and were not secondary to shrinkage of dendrites. NMDA antagonists blocked both acquisition of the conditioned response and the spine density changes, further indicating that the spine alterations were correlated with learning and not due to some other variable affiliated with the task. Synapse

²¹ Long-term potentiation (LTP), triggered by a strong repetitive stimulus or tetanus, is a physiological change in response thought to reproduce learning-related changes in synaptic efficacy.

²² Synaptogenesis, the formation of new contact between neurons, has been more difficult to demonstrate than has change in synapse and spine morphology.

and spine density changes are not limited to associative learning. In a study of operant conditioning, chicks showed increased spine densities in the dorsal hippocampus at 24 h and 48 h after avoidance conditioning [128]. Synaptic change clearly occurs with learning.

A number of studies have further addressed whether these kinds of learning-related plastic changes are permanent, and the consensus is that they are not. O'Malley and colleagues [93, 94] have shown that increases in spine density in the dentate gyrus of hippocampus following both avoidance training and spatial learning on a water maze task are transient. By 72 h after training, spine densities revert to control levels. Transient changes in spine densities were also found in the chick brain following a passive avoidance task. Different brain regions showed increases in spine density at different time points. Olfactory areas in the chick brain showed significantly heightened spine densities at 24 h after training, but not at 48 h [59]. On the other hand, the hyperstriatum of the chick brain showed increased spine densities at 1 h, but not at 24 h, following training [24]. If there is no net change in synapse or spine density that endures well after learning occurred, how is learning-related information permanently stored? Several ideas have emerged that will be discussed next.

Are there Plastic Versus Stable Spines?

In order to deal with the dilemma of permanent memory storage, several researchers have hypothesized that some spines are temporary and some permanent [63, 82, 115]. According to one view (see Kasai et al., [63]), spines with large heads and possessing many AMPA receptors are stable and represent permanent memory storage, while learning is represented in small mobile spines that continuously form and retract. These smaller spines may be "silent" under physiological conditions, but nonetheless represent a "reserve" of spines, which can change into large spines if sufficient conditions are met.

Although stable populations of large spines have been demonstrated in mouse neocortex,²³ spines are generally thought to be in a constant state of flux and this is regulated by the various neurotransmitter receptors. Glutamate receptors play a prominent role. AMPA receptor activation is important in spine stability, and constant levels of activation within the physiological range must be maintained [29]. The range of optimal activity is tightly regulated, since too high concentrations of glutamate produce a near complete retraction of dendrite spines and a significant reduction of F-actin in a process mediated by calcineurin, NMDA receptor and AMPA receptor [44]. Excessive AMPA receptor activation in developing neurons has also been shown to induce neurite retraction [101]. Similarly, kainate applied to neurites in cell culture produces a retraction of spines [49].

²³ Layer V pyramidal cells in visual cortex were tracked for 10 months and spines were found to be remarkably stable [41].

Although spines retract, they also reappear, often in their original locations. Curiously, actin-depolymerizing agents do not prevent the reappearance of spines in their original locations [49]. That spines are capable of reappearing in their original position without the aid of actin argues strongly against information (i.e. memory) being permanently stored within those actin-rich spines. Another piece of contradictory evidence to the hypothesis that memory is stored in stable actin filaments is that 85% of actin in spines is dynamic, not stable [118]. These findings suggest that the physical basis of memory must be stored elsewhere, perhaps in the subsynaptic zone that gives rise to the spine. Microtubules in the subsynaptic space are capable of storing important information concerning where spines were originally located, and to use that information to determine which synapses will persist or reappear in the event of dynamic retraction. Hence microtubules are satisfactory sites to store memory.

Transmembrane proteins may also play a role, along with microtubules. Neural cell adhesion molecules (nCAMs) are transmembrane glycoproteins that link the presynaptic and postsynaptic membranes. Both nCAM-140 and nCAM-180 bind α -tubulin, β -tubulin and α -actinin 1 [15]. Microtubules, in receiving neurons, may connect to microtubules in the axon terminals providing them inputs via a contact matrix composed of nCAMs and other proteins that traverse the synapse. This microtubule-nCAM-microtubule linkage may help guide new spines to form again in their original positions.

Another way to identify the site of memory storage is to pinpoint sites of high metabolic activity. The process of memory consolidation requires high metabolic activity, although once material is learned, the brain expends much less metabolic energy when using that information. In this regard, ATPase is sequestered in high concentrations in the subsynaptic space, not in the synapse [20]. At this location, ATPase is able to tightly regulate ATP levels, which are known to affect the viscosity and polymerization of the cytoskeleton [132].

A critical part of a microtubule mechanism of learning is that the microtubule assumes the role of an "active" participant, transiently crosslinking with actin filaments to selectively join subsynaptic zones together into a neuron-wide network. Self-organizing microtubules inside dendrites are able to polymerize and depolymerize and thereby search for subsynaptic sites with which they can connect to actin filaments²⁴. During learning, glutamate, acetylcholine and monoamines are likely to initiate microtubule polymerization or depolymerization²⁵. However, when there is no stimulus, such as

²⁴ By comparison, microtubules orchestrate mitosis by a search-and-contact strategy involving polymerization/depolymerization cycles, and by a third step that does not involve physical contact [121].

²⁵ These neurotransmitters affect microtubule polymerization through postsynaptic receptors that activate PI-PLC, and subsequently PKC and CaMK II, which in turn phosphorylate MAP2.

during spontaneous recall of a memory, a biophysical state of the microtubule must self-induce a polymerization/depolymerization cycle. Microtubules have long been recognized as self-initiating mechanical operators in the cell²⁶. Given the complexity of microtubule networks, a single neuron gains the computing capability of a large neuronal ensemble. As will be discussed towards the end of this chapter, microtubules may use various strategies when participating in brain-wide networks.

Dendrites are Comparatively More Stable than Spines: Microtubules Maintain Dendrite Integrity

Microtubules show greater stability than actin spines [62]. Although microtubules continuously lengthen and shorten *in vitro* and *in vivo*, they can also be very stable, even when the neuronal membrane is removed [123]. Several conditions contribute to the stability of microtubules. For one, crosslinking with MAP2 or tau markedly increases the structural stability of microtubules in dendrites [22, 48, 81]. Secondly, posttranslational modifications, such as de-tyrosination, polyglutamylation or acetylation, have the potential to enhance microtubule stability, not directly, but through increased binding to MAPs or neurofilaments [137]. Detyrosinated microtubules become associated with neurofilaments possibly via kinesin interactions [69]. Detyrosinated microtubules are exceptionally resistant to polymerization; tubulin dimers must first be fragmented in the presence of calcium before significant polymerization will occur [60]. Acetylation has been shown to stabilize microtubules in the subsynaptic region of the neuromuscular junction, which in turn, stabilizes acetylcholine receptors [61]. Posttranslational modifications of microtubules also perform a unique role in living cells. There is an abrupt change in tyrosinated tubulins and acetylated tubulins such that the composition of a microtubule gives a “snapshot” of its growth history [11]. This poses an exciting means to store information about when microtubule networks were reorganized.

Thirdly, attachment of the microtubule, especially at its ends, contributes to microtubule stability. There are a number of plus-end binding proteins (+TIPs) that stabilize the plus ends of microtubules in dendrites [119]. These +TIPs move bidirectionally along microtubules in the initial segment of dendrites, while movement is mostly outward in distal dendrites and axons. The initial segments of neurites are exceptionally stable [114]. The various means by which microtubules can be stabilized make these polymers good candidates for permanently encoding memory. These mechanisms of stabilization would favor the preservation of the overall structure during ongoing protein turnover.

²⁶ Microtubules self-assemble during cell mitosis.

3.3.2 Microtubules and MAPs in Dendrites Play a Critical Role in Memory

Microtubules and MAPs have a clear involvement in learning and memory. We showed MAP2 proteolytic breakdown occurs with memory consolidation [144–146]. We demonstrated MAP2 breakdown in two ways. First, we found increases in immunocytochemical staining in repeated experiments, using a panel of specific monoclonal antibodies. Increases in antibody binding are known to occur following proteolytic breakdown of cytoskeletal proteins because the breakdown exposes more antibody binding sites. The immunohistochemical method is superior at showing where breakdown occurs at the cellular level, but is not definitive at proving proteolysis. For this reason we also ran immunoblots. In these immunoblot experiments, we found an increased presence of breakdown products; results that proved proteolysis occurred.

The extent of MAP2 proteolysis spanned the entire cell body and main dendrites of the MAP-rich cells of those cortical area involved in the learning task. For the hippocampus-dependent task (learning an association between context and shock), MAP2 changes were found in CA1 and CA2 of the hippocampus. For the hippocampus-independent task (learning an association between tone and shock), MAP2 changes occurred in areas of the auditory cortex. In both these experiments, learning correlated with a simultaneous breakdown of MAP2 throughout an entire functional unit of the cerebral cortex (Fig. 3.7).

In our experiments, individual areas of cortex in each animal studied showed a different intensity of MAP2 staining. This, along with the learning-related changes, indicated that the modular variability of MAP2 staining reflects the recent experiences of the animal. Another group found a similar kind of modular variability for MAP2 and PKC immunostaining following passive avoidance training [130].

What induces this pattern of MAP2 proteolysis? Since MAP2 bound to microtubules is more susceptible to proteolysis [38], it is likely that the staining patterns we observed largely reflected the proteolysis of intact microtubules²⁷. Proteolysis is triggered by increased levels of Ca^{2+} , which activate calpain, which in turn degrade MAP2. Buddle and colleagues [14] have shown that calpain-induced MAP2 proteolysis during oxygen-glucose deprivation is triggered by NMDA-receptor activation. Acetylcholine M1, M3 and M5; serotonin 5-HT₂; α 1-adrenergic and mGlu1 and mGlu5 receptors are also candidates for activating calpain though PI-PLC-stimulated release of endogenous Ca^{2+} . Destruction of cholinergic fibers innervating the cerebral cortex results in decreased MAP2 staining, indicating that a hypocholinergic state may reduce intact MAP2, reduce the proteolytic breakdown of MAP2 or both [147].

²⁷ We also observed a small, but significant breakdown of tubulin [145]. This probably reflected the breakdown of unbound tubulin; however, it could have been tubulin that dissociated from intact microtubules.

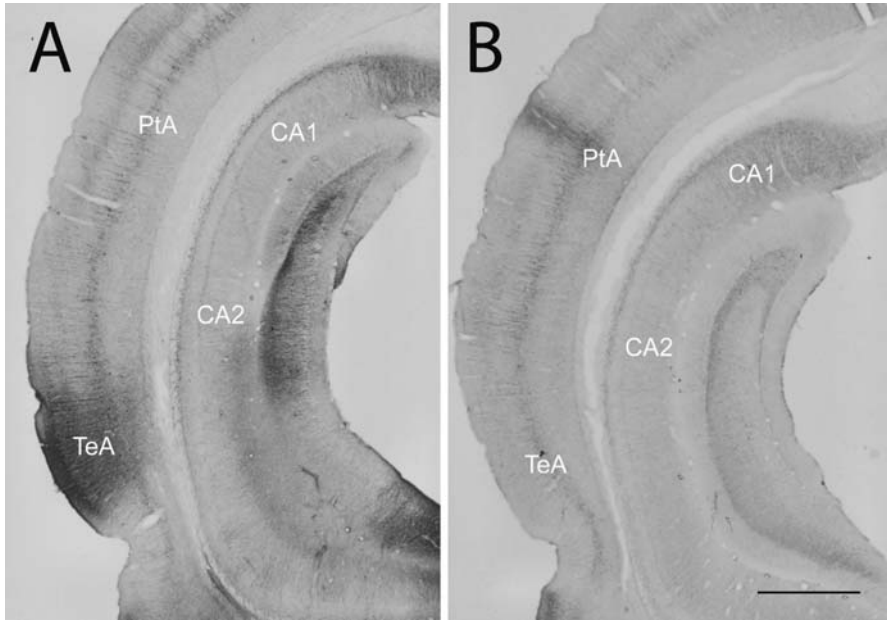


Fig. 3.7. Photomicrographs showing the distribution of intensified immunostaining for MAP2 in a module of temporal association cortex (TeA) of an animal trained on a fear-conditioning task (frame A) compared to that of a naive animal (frame B). Conversely, a more slender module in the parietal association cortex (PtA) is more intensely stained in the naive control animal than in the fear-conditioned animal. Sectors CA1 and CA2 of the hippocampus are lightly stained in both animals. Scale bar equals 1 mm

Cholinergic fibers arising from the basal forebrain have terminal fields that are approximately the same size as cortical areas or modules in the rat²⁸, further implicating acetylcholine as playing a role in the MAP2 response with learning. If our results had revealed MAP2 breakdown mainly in the fine distal dendrites, then a change in dendrite branch geometry would be the logical interpretation²⁹. In our observations, however, a significant amount of MAP2 degradation occurred in the cell body and in the proximal dendrites. This implies that a reconstruction of intradendritic microtubule networks occurred, since new branches are rarely added to the thick proximal dendrites or to the cell body. Hypothetically, updated microtubule networks could serve many

²⁸ Cholinergic fibers terminate in fields of 1–2 mm², whereas glutamate fibers terminate in vertical columns of 50 μ m, and noradrenergic, dopaminergic, and serotonergic fibers diffusely innervate widespread expanses of cerebral cortex [148].

²⁹ Breakdown of MAP2 and subsequent phosphorylation may increase branching, since increases in the MAP2 phosphorylation state correlate with increases in dendrite branching [6]. Increased dendrite branching has been observed with certain types of learning (e. g., Withers and Greenough [140]; Greenough et al. [39]).

purposes. Newly formed intradendritic networks could connect different subsynaptic zones together, thereby storing information about the spatial and temporal relationships of activity at those synapses.

Plus ends of microtubules are primary sites of polymerization, and these sites are distributed throughout the neuron. Local polymerization/depolymerization is instrumental to updating the microtubule network in all parts of the neuron³⁰. When plus ends of microtubules near subsynaptic zones are affected by synaptic activation via glutamate neurotransmission, this triggers a localized response. The additional presence of a neuromodulator, such as acetylcholine, will promote widespread effects along the dendrite shaft. Thus, the coactivation of glutamate, along with acetylcholine or another neuromodulator, is likely to produce the greatest change to the intradendritic network of microtubules, in terms of both localized and widespread effects.

Relatively stable MAP2 bridges near the minus ends of microtubules may be most critical for linking together different coactivated subsynaptic zones and providing a possible means for long-term storage by microtubules (refer back to Fig. 3.6). Accordingly, the microtubule network, with its various MAP2 linkages, could serve as a coincidence detector for multiple synapses within the same neuron. More than merely linking two coactivated subsynaptic zones, layers of microtubules could build complex parallel networks representing coactivation patterns, using a reasonably small number of relatively stable MAP2 bridges³¹. Microtubules could, furthermore, weight synaptic inputs equally or unequally depending on their lengths, or series of microtubules of different lengths could store temporal sequences of activation. The possibilities are almost endless.

The various modes by which microtubules may store information could account for how microtubules might determine where spines are formed (and where they would reappear following retraction). There is preliminary experimental evidence that spine motility may depend on microtubules. The vinca alkaloid, vinpocetine, increases spine motility, apparently because of its effects on microtubules [74].

Reorganization of microtubule networks and their interactions with actin filaments could also increase (or decrease) the efficiency of transporting intact synaptic protein and mRNA to certain synapses and spines. In this manner, microtubule-actin networks could initiate or maintain potentiated or depotentiated synaptic activity. Evidence in support of this is found in long-term potentiation (LTP) relying upon the transport of AMPA receptors along microtubules and actin filaments [65, 66]. Also, LTP increases a number of presynaptic and postsynaptic proteins, including F-actin [5]. Maintenance of

³⁰ Stepanova et al. [119] used end-binding protein 3 (EB3) to show local microtubule polymerization.

³¹ Stable MAP2 bridges would be expected to turn over at slower rate, allowing for replacement with new MAP2 bridges that preserved the intradendritic pattern of connectivity.

the late phase of LTP depends upon F-actin [31]. Involvement of microtubules is also implicated during particular phases of LTP. The mRNAs for CaMK II and MAP2 are increased during the induction of LTP [105]. During the intermediate phase of LTP, expression of CaMK II and MAP2 is increased by calcium and PKA, respectively [3].

Microtubule networks also have a means to “date” when groups of new synapses are formed or changed. Posttranslational modifications such as dephosphorylation reflect the age of microtubules. Synapses that are formed or modified at the same time will be associated with subsynaptic microtubules having the same “age”.

An Anchoring Site for PKA on the N-Terminus of MAP2 is Necessary for Learning

As emphasized earlier, MAP2 is not merely a structural protein. MAP2 also serves as a PKA-anchoring protein and it binds regulatory subunits RII α and RII β of PKA. This aspect of MAP2 has been studied in knockout mice. Mice bred homozygous for a genetic knockout of MAP2 (MAP2 $^{-/-}$) show a reduction of microtubule density and dendritic length, as well as decreased PKA expression [48]. The RII α , RII β and catalytic subunits of PKA are decreased in MAP2 $^{-/-}$ mice as compared to wild-type mice, whereas CaMK II expression in MAP2 $^{-/-}$ mice is about the same as in control mice. Thus, it appears that MAP2 tethers PKA to the microtubule, until PKA release into the cytoplasm is stimulated by cAMP.

The regulatory subunit RII β of PKA is located on the N-terminus of MAP2. Deletions of the N-terminal region of MAP2 were made in another set of transgenic mice, and then these mice were studied for effects on behavior [64]. These transgenics were impaired on a fear-conditioning task to context, but not with fear conditioning to tone³². Moreover, cAMP-dependent phosphorylation of MAP2 was significantly diminished in these transgenics. In heterozygous mice, cAMP-dependent phosphorylation was decreased by 90% and 80%, respectively, for truncated and full-length MAP2. Decreased phosphorylation for full-length microtubules was unexpected. The explanation given by these authors is that MAP2 forms multimolecular complexes and then transphosphorylates tubulin binding domains on neighboring MAP2 molecules. This transphosphorylation could lead to the dissociation of MAP2 and those microtubules. This result also indicates bridges linking microtubules in dendrites are likely to be composed of two overlapping MAP2 sidearms³³.

The conclusion of these studies is that MAP2 acting as an anchoring protein plays an essential role in learning. Additionally, MAP2 is part of

³² Conditioning to tone served as a control.

³³ Such an explanation would reconcile the impression in electron micrographs that MAP2 seamlessly links neighboring microtubules, yet only the C-terminal region of MAP2 has a binding site for tubulin.

a multimeric complex that regulates its own phosphorylation state and that of neighboring components of the cytoskeleton. The end result can affect the MAP2–microtubule linkages, the density of microtubules and microtubule length.

Microtubule Transport by Kinesin Participates in Learning

There are many kinesins. KIF17 belongs to the kinesin superfamily and it specializes in transporting NMDA receptor subunit NR2B in neurons. Guillard et al. [42] performed double-immunohistochemical analyses to determine the fate of NR2B subunits in hippocampal neurons. From their colocalization studies, they determined that KIF17 was responsible for transporting NR2B to synaptic sites labeled by PSD-95 or synaptophysin. Near to the synaptic site, NR2B detached from kinesin and colocalized with PSD-95 or synaptophysin. A separate study from the same laboratory showed that transgenic mice overexpressing KIF17 had improved working and spatial memory [142]. These authors propose a repeating cycle based on their results. The cycle starts with increased transport of NR2B in dendrites, which increases Ca^{2+} ion flux, and then increases in PKA and CaMK II. Next, the transcription factor CREB is activated, which increases mRNA of NR2B, and then the cycle repeats.

Polyribosomes and mRNAs for signal-transduction molecules critically involved in learning and memory have been found in dendrites [120]. These include mRNA for NMDA receptor, CaMK II, and MAP2. Local translation of these proteins within the spine is made possible by shifts of polyribosomes from dendrite shafts to spines following tetanic stimulation producing LTP [95]. These shifts among ribosomes further implicate transport as critical to learning. Other evidence suggests that kinesin plays a role in learning and plasticity. We have studied KIF17, NR2B and PSD-95 in rats for one week following training on the contextual fear task. In hippocampus CA1, contextual fear training decreased the binding of NR2 to kinesin and increased NR2 binding to PSD-95 (see Fig. 3.8).

Our results are consistent with the notion that brain reorganization following learning stabilizes increased levels of receptors bound to the postsynaptic density and decreases further transport. Since there is generally an inverse relationship between microtubule stability and transport, this would be consistent with increased microtubule stability at one week after training. Experiments done on MAP2, kinesin and learning can be summarized into an overall scheme that places microtubules in a central position during learning and memory (see Fig. 3.9).

Step 1. There is coactivation of acetylcholine and glutamate, as one would expect when a stimulus was receiving attention. Metabotropic receptors (e.g., muscarinic) increase phosphorylation of MAP2 and decrease its binding to microtubules.

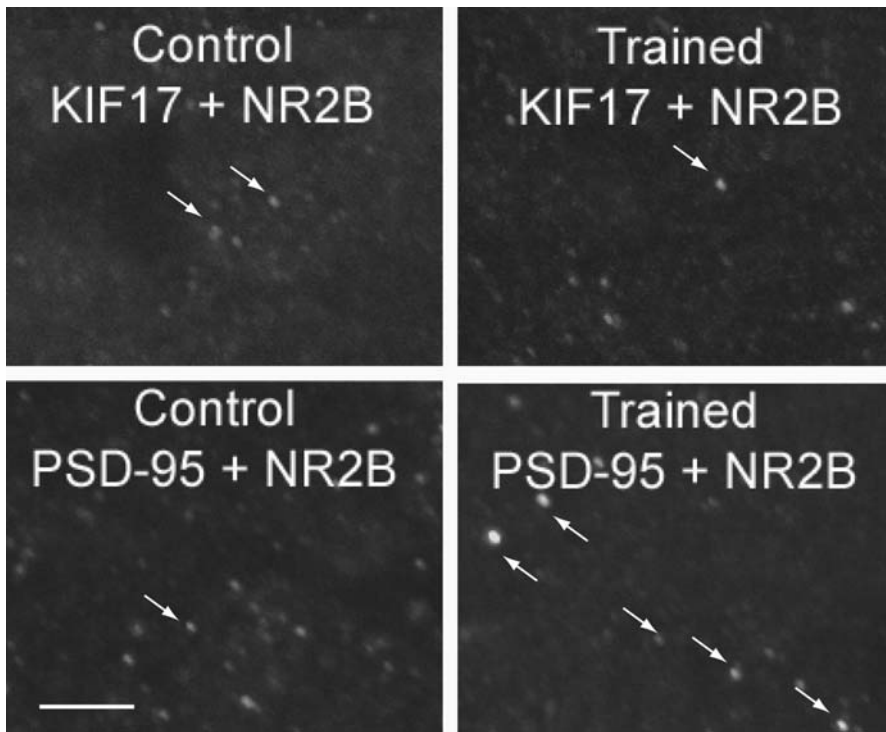


Fig. 3.8. Photomicrographs of hippocampal tissue from animals trained on a fear-conditioning task or naïve controls. *Top panels:* Kinesin, KIF17, is immunostained with fluorescein (illustrated as green puncta) and NR2B is immunostained with Texas red (illustrated in red). White arrows point to overlap between KIF17 and NR2B shown as yellow. *Bottom panels:* Postsynaptic density protein, PSD-95, is immunostained with Texas red (illustrated as red puncta) and NR2B is immunostained with fluorescein (illustrated as green puncta). White arrows point to overlap between PSD-95 and NR2B shown as yellow. Scale bar equals 10 μm

Step 2. Increased transport of protein, for example, NMDA receptor, would be expected as a result of decreased MAP2 binding, which would free up binding sites for kinesin. This is expected because MAP2 and kinesin bind to the same general vicinity on the microtubule.

Step 3. Upregulation of NMDA receptors would be expected as a result of increased transport of NMDA receptor subunits. This would also increase the influx of calcium ions, which would increase calpain.

Step 4. Proteolysis of MAP2 and tubulin leading to a dissociation of microtubule bridges and microtubules would be expected as a result of increased calpain.

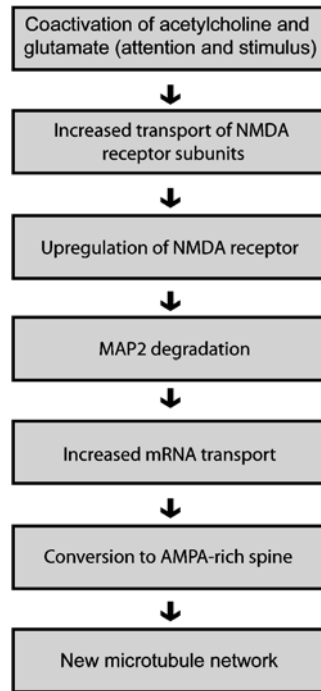


Fig. 3.9. The seven steps outlining the role of microtubules in memory. See text for elaboration

Step 5. Following MAP2 proteolysis, rebuilding would include increased transport of mRNA for NMDA, CaMK II and MAP2, followed by local translation of these proteins essential to learning.

Step 6. More rebuilding would include the conversion of small spines into larger, AMPA-rich spines. These large AMPA-rich spines would be maintained at their increased size by regulating transport rates of AMPA receptors. Microtubule networks would be responsible for determining spine size and receptor content.

Step 7. A new microtubule network would be formed as a result of extensive reorganization. Consequently, microtubules would direct kinesin-mediated transport to the various subsynaptic zones throughout the neuron.

Spatial relationships between synapses could be represented by MAP2 bridges between the minus ends of two or more microtubules having plus ends reaching the subsynaptic zones of those synapses to be linked. MAP2 bridges near the minus ends are emphasized here because these linkages are relatively stable³⁴. Temporal relationships could be similarly represented, but

³⁴ Microtubule polymerization and binding to MAP2 are stable near the minus ends due to posttranslational modifications, such as detyrosination, and linkages to neurofilaments.

with different lengths of microtubules linking subsynaptic zones such that activity could be reproduced in particular sequences. Memories formed at the same time could be marked by similar “dates” indicated by similar posttranslational modification patterns. These types of modifications, among others, would represent long-term memory storage.

3.3.3 Microtubules Influence Synaptic Efficacy

Once memory is stored in newly stabilized microtubules, how is that stored information able to affect neural processing? There are a number of ways microtubules could affect electrophysiological events, such as membrane potentials and action potentials. For one, by determining the placement of synapses and their size, synaptic activity will be altered in the long term. Secondly, by altering transport rates of proteins and mRNA to synapses, microtubules will affect membrane potentials within minutes to hours. Thirdly, the biophysical state of the microtubule has the potential to actively control the synapse and the neuronal membrane directly, acting within milliseconds.

Although the integrity of ion channels does not appear to depend on the cytoskeleton [86], microtubules directly regulate or modulate some gated ion channels. For example, cytoskeletal proteins regulate the GABA_A-gated channel [138]. Stabilization of microtubules appears to directly block muscarinic acetylcholine receptors from inactivating the L-type calcium current [79]³⁵. Polymerization state of microtubules affects the function of glycine receptors [23]. Microtubules have also been shown to abruptly shift the properties of sodium channels [113] and are necessary for proper functioning of sodium channels in squid axon [80].

The ability of microtubules to commence electrophysiological activity is an entirely feasible proposal. Microtubules self-induce polymerization/depolymerization cycles during cell mitosis, and may do the same in order to initiate electrophysiological responses. Self-initiated changes in biophysical properties affecting the binding and unbinding between microtubules and MAPs may also influence local ion-channel activity. Any model of memory must address the issue of how synapses are activated in the absence of direct sensory activation. Models such as potentiation and depotentiation describe the nature of electrophysiological responses *elicited by stimuli* following conditioning, but not how such cellular responses are initiated in the *absence of stimuli*.

3.4 Consciousness

Consciousness, memory and attention are interrelated psychological phenomena, although key differences are found between them. Surface recordings

³⁵ This was found in guinea-pig myocytes.

from human subjects show synchronous oscillations of different frequencies typify memory in contrast with attention, and that consciousness is correlated with widespread synchronous oscillations³⁶. Nonetheless, it is difficult, if not impossible, to attempt to explain consciousness without addressing the psychological phenomena that it makes possible. An important concept often neglected is that conscious activity is action. Consciousness is covert behavior, the internalized brain function that is the counterpart of overt, observable behavior.

In order to make pragmatic use of information stored in some types of memory³⁷, that information must be brought to consciousness. Psychologists refer to this process as memory retrieval. A possible brain mechanism for memory outlined in the previous section of this chapter assigns an “actor” role to microtubules in the brain. This mechanism is also able to account for the action component of consciousness.

3.4.1 Attention: The Spotlight of Consciousness

Attention is a deliberate action whereby one directs conscious operations to a particular sensory stimulus, to a memory, or to a complex amalgam of both stimulus and memory. Sensation, attention and arousal are mediated through different neurotransmitter pathways, each of which will affect brain microtubules in different ways.

Glutamate neurotransmission transmits sensory data all along the sensory pathways. A sensory pathway begins in a sensory organ (i.e. the retina, cochlea, vestibular organ, skin, taste buds or nasal epithelium), usually makes synaptic contact in the thalamus, then relays to a primary sensory cortex, and then through a series of corticocortical circuits reaches higher association cortex. The glutamate system is topographically organized (e.g., visuotopic, tonotopic or somatotopic) such that the location of the visual field, the cochlea, or the body onto which stimuli are mapped is preserved at each relay. The sensory systems organize information from external sources in a point-to-point scheme. By analogy, it is like every person living in New York City being in phone or email contact with one person in Los Angeles who lives at the same north-south and east-west placement in the corresponding city. This system is “online”; new information is continually coming in.

Global systems of neurons utilize the neurotransmitters acetylcholine and monoamines: norepinephrine, serotonin and dopamine³⁸. These cellular aggregates receive sensory collateral branches from axons en route to the cere-

³⁶ Memory is related to oscillations in the theta and gamma range, whereas attention is related to alpha and gamma [135].

³⁷ Memory that reaches consciousness is called explicit or declarative.

³⁸ These groups of neurons form interconnected cellular aggregates throughout the hindbrain and basal forebrain enabling a high degree of crosstalk between constituent neurons [148].

bral cortex. In this manner, these neurons obtain a general idea of the overall pattern of incoming inputs and then base their outputs on those assessments. Cholinergic systems contribute to selective attention, whereas norepinephrine, serotonin and dopamine systems contribute to general arousal or vigilance. Cholinergic neurons from the basal forebrain are able to contribute to selective attention because they innervate discrete areas of cortex of a single modality measuring 1–2 mm². Noradrenergic, serotonergic and dopamine neurons in the brainstem innervate the cortex less restrictively. A single noradrenergic fiber, for example, may provide inputs to many different types of cortex, as these fibers are long and have many release sites along their paths.

Taking into consideration that attention is a deliberate act, what neurochemical component of the neural circuit could be responsible for it? The glutamatergic synapse per se cannot be the source of attention. Sensory inputs in some cases do reach consciousness, but often they do *not* reach consciousness. What part of the glutamate synapse would decide, and on what basis? These synapses are passive, responding to inputs. Even in the presence of enhanced synaptic efficacy (e. g., LTP), how would such potentiation lead to attention and consciousness? Going back to the analogy of one person in New York City being in direct contact with one similarly placed person in Los Angeles, we might view increased synaptic efficacy as an increase in information flow between selected persons in this one-to-one connected network. Would selectively increasing information flow attain an increased level of attention or reach consciousness? No, not in this one-to-one connected network, which is not too far from the way cortical systems are organized. A mechanism is needed that accesses the overall pattern of inputs and compares these patterns to input patterns occurring in the remembered past and input patterns anticipated in the future.

Cholinergic and monoaminergic inputs seem closer candidates for this role of accessing the overall pattern of inputs, insofar as these systems sample sensory activity en route to the cerebral cortex. On the other hand, the information these systems relay cannot be precise enough, since their terminal fields are diffusely distributed to cortical modules or to the entire cortex. These inputs are necessary, but not sufficient for attention and arousal. Activation by glutamate, acetylcholine and norepinephrine are obligatory steps for a stimulus to receive attention. But all these synapses are passive; none has the “actor” status needed to decide whether to deliberately arouse consciousness or to select content.

Self-organizing microtubule networks inside dendrites do have the “actor” status needed to deliberately direct attention and consciousness. Because microtubules can polymerize and depolymerize, they can search for and activate particular subsynaptic sites. Although release of glutamate, acetylcholine and monoamines is obligatory for full conscious awareness, this behavioral state must also rely on the biophysical state of the microtubule network. Just as phosphorylation of MAP2 and its subsequent binding can affect the biophys-

ical state of the microtubule, the biophysical state of the microtubule can affect the binding of MAP2. As a specific example, the biophysical state of the microtubule can induce the release PKA through effects on the regulatory subunit RII β of PKA. Bidirectional effects on PKA-mediated transphosphorylation of neighboring MAP2 causes the “zipping” and “unzipping” between neighboring microtubules, leading to subsequent increases and decreases in the rate of kinesin-mediated transport along microtubules³⁹. Regulation of transport is necessary to maintain large spines, which are laden with many AMPA receptors.

These, among other effects, would enable the microtubule network to direct attention and consciousness. Accordingly, the psychical sensation of attention could not possibly be associated with synaptic activity per se, since those activities occur both in the presence and in the absence of attention. Instead, the psychical sensation of directed attention and consciousness could well be associated with the biophysical state of microtubules that regulates (1) polymerization and depolymerization cycles linking synapses and (2) transport to active synapses.

3.4.2 Waking, Sleeping and Dreaming: Different Levels of Consciousness

Another way to look at consciousness is to see it as part of the sleep–wakefulness continuum. When a subject is awake, the electroencephalogram (EEG) is a complex mixture of fast waves (i.e. beta, alpha, gamma waves). When a subject falls into dreamless sleep (non-REM⁴⁰), the EEG pattern slows down markedly, each deeper step containing fewer theta waves and more delta waves. When a subject experiences REM or dream sleep, the EEG looks more like that of an awake person than that of a person during non-REM sleep. It is not known what provides the basis for the content of dreams.

Neurotransmitter levels change over the sleep–wakefulness cycle and bilateral damage to brainstem areas containing cholinergic, noradrenergic and serotonergic neurons results in coma [99]. Acetylcholine, norepinephrine and serotonin levels are high during wakefulness, low during non-REM sleep; and during REM sleep, acetylcholine is very high and norepinephrine and serotonin levels are very low [53, 55]. This pattern suggests norepinephrine

³⁹ When adjacent microtubules are linked by antiparallel MAP2s, transphosphorylation might be expected to repel these MAP2s leading to the “unzipping” of the two microtubules and thereby removing any physical block impeding kinesin binding. Although kinesin binding is not directly affected by MAP2s as it can travel on protofilaments not bound to MAP2s, attached cargo and other complexed proteins will affect the processivity of kinesin in a limited space.

⁴⁰ Sleep is characterized by periods of rapid eye movements (REM) and periods without rapid eye movements (non-REM). REM sleep corresponds to dream sleep most of the time.

and serotonin are critical in linking consciousness to incoming stimuli. These monoamine neurotransmitters often bind to receptors activating the second messenger PKA, suggesting that high levels of cAMP and PKA are important to wakeful consciousness. But during dream sleep, norepinephrine and serotonin levels sink very low. Since dreaming is a kind of conscious activity – consisting of visual imagery, thinking and emotion – acetylcholine might be particularly responsible for those aspects of consciousness reflecting inner experience. As mentioned earlier, however, cholinergic terminals are too diffusely organized to account for the specific content of inner experience, and the same argument would hold for the specific content of dreams. Perhaps effects of the combined release of acetylcholine and glutamate could account for the content of dreams. But during dreaming, activation of glutamate synapses must be entirely random. Also, combining neurotransmitters does not solve the problem concerning self-organization.

One is left with the only possible conclusion being that neurotransmitter release per se cannot be responsible for consciousness. Since a tenable scenario is that a biophysical state in the microtubule network is responsible, corelease of glutamate and acetylcholine could facilitate dream-state consciousness through enabling effects on microtubules. In this case, the psychological sensation of consciousness would be related to the biophysical state of the microtubule network, as described earlier.

The basis of anesthetic action demonstrates that a relationship between the cytoskeleton and consciousness does exist. Many of the cytoskeletal actions thought to be critical to memory are also blocked by anesthetics. Specific examples include: local anesthetics inhibiting kinesin motility [84], volatile anesthetics interfering with actin-based motility in dendritic spines [62] and volatile anesthetics interfering with MAP2 degradation in frontoparietal cortex and hippocampus [122]. These studies support a link between the cytoskeleton, memory and consciousness.

3.4.3 Mental Force to Think and Act

As a general rule, the more involved that analyses of cortical responses get, the more it becomes clear that to understand the simplest of cellular responses, one needs to simultaneously understand brain-wide activity. Consider a simple perceptual act, that of seeing a simple contour. Complex cells in the primary visual cortex respond to salient contours embedded in complex backgrounds [77]. Long-range horizontal connections provide a rudimentary basis for contour saliency; however, top-down influences and learning effects are needed to explain how this, one of the simplest percept, comes to be realized.

So what does one attempt to understand first: the complex multisensory processes or the simple early sensory responses? As stated earlier, there are a few known multisensory regions. These brain regions having multisensory

connections appear to use those connections for multisensory tasks. A recent fMRI study, for example, found a small region in the posterior intraparietal sulcus was activated specifically for the crossmodality component of a combined visual-auditory task [107]. One might not expect crossmodality responses during early processing in primary and secondary sensory cortices, yet many examples of these kinds of responses appear in the literature. Neurons in secondary auditory cortex tuned to precise frequencies of sound also respond weakly to somatosensory input [30]. Eye position affects processing in primary auditory cortex [136]. Visual stimuli enhance tactile acuity through modulation of responses in the primary and secondary somatosensory cortex [124]. There are crossmodal links in selective attention directed toward visual, auditory, somatosensory and olfactory stimuli [116]. One study found elevated activity on fMRI scans in primary, secondary and association visual areas when congenitally blind subjects read Braille, but not when their fingers were merely touched [35]. This last study proves that crossmodality responses are not driven through the growth of plastic connections between somatosensory and visual areas. Cortical plasticity, in terms of new axon growth, cannot account for how these effects rapidly appear under certain conditions and then disappear under other conditions instantly.

Crossmodal influences may occur through feedback connections from multisensory cortex. Sparse anatomical links from multisensory association cortex have been traced to primary and secondary visual cortex [27, 106]. Critical questions remain unanswered, however: Are the sparse connections currently known sufficient for the multitude of crossmodal tasks? How do these areas decide when to transmit information backward, and to where? Since most cognitive tasks we perform have multisensory components, it appears that: (1) the same few multisensory regions are involved in many tasks, (2) many anatomical connections remain undiscovered or (3) some kind of brain interactions other than corticocortical connections are additionally at play⁴¹.

Some researchers have acknowledged the lack of critical corticocortical connections implied by certain visuomotor tasks and have looked to subcortical connections [36]. Many, if not most, subcortical sites are also segregated according to sensory modality⁴². How is it then that we can guide complex voluntary motor acts for the first time? Human subjects are known to impart their own personal handwriting style when using their toe to write a message

⁴¹ Corticocortical connections link different cortical areas preserving much of the original sensory topography; thus, the anatomical scheme continues to be a point-to-point mapping with limited opportunity to synthesize and integrate data. Modulatory brain regions (e.g., the basal forebrain and locus ceruleus) are anatomically better suited to integrate and synthesize multisensory responses in crossmodal selective attention because their projections onto cortex are more diffuse. Projections from the locus ceruleus are more diffuse than those from the basal forebrain.

⁴² Edge et al. [25] found evidence for somatosensory and visual convergence in parts of the cerebellum.

in the sand⁴³. How much plasticity is needed to produce connections that are largely or completely absent before attempting such a behavior for the first time?

Perhaps a well-developed sensorimotor imagination is enough. Motor imagery is essential for simple movements. Recently, it has been shown that the motor cortex is activated as if sensory input had occurred in anticipation of a motor activity [91]. This shows that we experience sensory feedback before it occurs. Nonhuman primates have similar sensorimotor skills to ours. Monkeys can learn to produce coordinated activity in large ensembles of frontoparietal cortex in order to control robotic arms through implanted arrays of microwires [17]. Monkeys learn this task without any somatosensory feedback⁴⁴; it is as if these animals can imagine that feedback. This “imagined” somatosensory feedback and the enigmatic “will to move” are mainly driving the neural activity that moves the robotic arm.

What provides the “mental force” needed to begin a movement or to drive thought? Microtubule networks and motor proteins fit the bill because of their mechanical nature and their interactions with actin filaments in linking synapses. Through self-induced polymerization/depolymerization cycles, binding and unbinding to MAP2 and kinesin, microtubules are capable of influencing membrane potential differences, in both the short-term and long-term ranges.

3.4.4 Consciousness, Memory and Microtubules

If memory is stored in microtubules, and if microtubules are the site of consciousness in the brain⁴⁵, then it is a simple case to relate memory and consciousness. Instead of using one hypothesis upon which to base another, however, I will argue that we can eliminate certain possible bases of consciousness if those are not suitable bases for memory storage.

Changes in synaptic strength undoubtedly play a role in encoding memory, but is this also the mode of long-term storage? The contemporary view is that long-term memory storage is the change within the synapse, for example, a thickening of the postsynaptic membrane, a change in spine shape or receptor density. The view I propose in this chapter is that the site of memory storage is not within the synapse, but in the subsynaptic zone, in the microtubules of the dendrite shaft. In addition to the data presented earlier, the logic to this proposal is that the synapse is a channel transmitting information, much like a radio receiver. To record information transmitted on a channel, a separate recording material, such as magnetic tape, is needed. If

⁴³ Based on studies of motor equivalence in handwriting [139].

⁴⁴ Monkeys keep their own arms still in this task and receive no direct feedback from the implanted device.

⁴⁵ See supporting arguments and outline of potential falsifying experiments in Hameroff [46] and Hameroff and Penrose [47].

one permanently altered the receiving channels, the device would not maintain fidelity; it would not be able to faithfully transmit audio information in the future. It would be analogous to having Beethoven played on the radio today sounding exactly like Bach, which was played yesterday. That is not the way memory systems work, nor could they. It would serve no adaptive function to perpetually alter our sensory input channels; it would prevent any real comparisons and any degree of perceptual constancy.

In this analogy, the receiving channel is the synapse or spine. Although change in select populations of synapses or spines is part of memory encoding, altered synapses and spines soon revert back to baseline. Thus, there appears to be a homeostatic mechanism keeping the parameters of synapses and spines within certain boundaries. On the other hand, memory consolidation involves massive reorganization of microtubule networks⁴⁶. Reorganization this extensive is unlikely to revert back to baseline. Permanence of long-lasting change in microtubule networks is also revealed by changes to overall dendrite structure and dendrite-branching patterns long known to occur with learning. The arbitrator of dendrite morphology is the microtubule.

Most contemporary models of consciousness posit synaptic activity as the primary basis of consciousness. Since low levels of synaptic firing occur even when a subject is unconscious, it is clear that consciousness cannot be synaptic activity per se. Synchronous activity has been suggested as the neural correlate of consciousness; however, this concept has its counterexamples. Slow-wave sleep, for example, is the stage when neuronal synchrony is highest. Thus, it has been proposed that synchrony in certain frequency ranges, for example, in the gamma frequency range, constitutes consciousness. Here we have the problem that temporal information would not be preserved in such a scheme. How would one be conscious of a particular pattern of auditory clicks, for example, if that temporal pattern of input was obscured by a different consciousness-related frequency? Another notion put forward is that consciousness emerges as a function of complexity as groups of neurons fire together in various patterns. This idea is laudable, but arguably incomplete: a necessary element is missing.

A microtubule basis of memory and consciousness is a departure from current doctrine in neuroscience, but it may have fewer conceptual flaws than widely perceived. It is not a great leap in intellect to suggest that the sub-synaptic zone, rather than the synapse, stores memory in the form of microtubule networks capable of self-organized polymerization/depolymerization cycles and transport, which in turn regulates synaptic activity. It is, furthermore, no monumental stretch of logic to postulate that the psychical sensation of consciousness is affiliated with the most directly relevant physical process, that being the biophysical state of the microtubule network.

⁴⁶ MAP2 degradation indicates widespread reorganization of microtubules with fear conditioning [144, 145].

3.5 Microtubules and Quantum Entanglement: A Possible Basis for Memory and Consciousness

Since the brain is not as widely interconnected as might be generally assumed, how is it that the brain acts as an integrated holistic system? How can activity at synapses on the same neuron or on different neurons become perfectly unified in order to represent a single idea or act? First, it is important to point out that being unified is not the same thing as being connected. During many perceptual tasks different parts of a whole are experienced as unified, but the parts do not necessarily *communicate* with one another. The left part of the visual field does not need to commute with the right part of the visual field; nonetheless, they must be coexperienced. The same is arguably true for sensorimotor tasks. Given this perspective, it becomes clear that unified experience has more to do with coactivation of synapses rather than connections between neurons. But if mere coactivation of synapses fully accounted for the experience of mental events, then this would greatly reduce the need for learning-related change in the nervous system. The first instance of coactivation would suffice. All one would need is to re-experience that initial unified activation pattern again⁴⁷. This illustrates a possible fundamental error in the current viewpoint that neural-network-type learning is responsible for cognition, since, after periods of enhanced synaptic efficacy, the theory still relies on coactivation of synapses for the representation of ideas. One is left no closer to understanding the physical basis of the mental representation itself. If, on the other hand, the subsynaptic zones of highlighted synapses are connected by systems of microtubules, then quantum entanglement among those microtubules *is* a possible solution to unification, a solution with a real physical basis. This physical representation can then, in turn, be unified by quantum entanglement with other unified ideas represented by the coactivation of other subsynaptic zones.

Since the brain is so complex, I suggest the following scheme of quantum entanglement occurring at five levels of neural interaction based on the proximity of tubulin or microtubules to one another. These levels of neural interaction have nothing to do with the physics of entanglement per se, but rather with how one might expect brain tissue to be differently affected by this physical phenomenon at each level of neural interaction. A key point is that a small amount of entanglement is predicted to link the minimum number of subsynaptic zones required at each level. Accordingly, neural events will be tightly or loosely coupled depending on their degree of physical interaction.

The first level of neural interaction would include quantum entanglement among tubulins of the same microtubule. As is the case for all molecules in

⁴⁷ A possible counterargument is that learning is needed to reduce synaptic activity since experience leads to less, not more, cortical activation. Nonetheless, this is a quantitative issue and has little to do with the qualitative nature of the physical basis of mentation.

neurons, tubulins exist in a crowded molecular environment and perturb other molecules as they undergo chemical and physical changes. Under appropriate circumstances, these effects are likely to cascade. This is especially true since tubulin molecules are linked into long microtubule polymers; a biophysical change, such as deformations in dimer-dimer interactions, may perturb tubulins for several millimeters along the microtubule [109]. Since this level of neural interaction is between closely interacting particles, the degree or density of entanglement is likely to be high. Thus, the subsynaptic zones beneath adjacent synapses could, in principle, exhibit strong entanglement. Entanglement of this type is possible in brain tissue, according to recent calculations⁴⁸.

The second level of neural interaction would involve entanglement between microtubules in the same neuron. MAP2, for example, does link neighboring microtubules together (or to actin filaments) and these linkages dynamically associate and dissociate. Each MAP2 has one microtubule-binding site; however, it is likely that pairs of MAP2-microtubule complexes interact physically or via transphosphorylation, a process that promotes the exchange of phosphoryl groups between the two antiparallel MAP2-microtubule complexes. Microtubules connected (or having been connected) through MAP2 linkages or having exchanged phosphoryl groups would appear capable of entanglement due to present (or previous) interactions among particles. This would be especially true at the minus ends of microtubules, which tend to be stably linked to neighboring microtubules by MAPs and neurofilaments. This degree of entanglement would be somewhat less than that of tubulins in the same microtubule, meaning there might be more quantum discord. This would enable each neuron to assume many different states, linking different subsynaptic zones in a variety of ways.

The third degree of neural interaction would involve entanglement between neurons in modules. Our data showing uniform MAP2 proteolysis within neurons throughout cortical modules could be interpreted to reflect simultaneous degradation occurring in coherent quantum systems of microtubules within cortical modules. Horizontal gap junctions link microtubules from separate neurons together, physically connecting the cytoplasm of pyramidal neurons, horizontal interneurons and glial cells. Connexin-43, a widely expressed gap-junction protein, also binds to tubulin [34]. We have previously proposed that these gap junctions could underlie microtubule-based quantum entanglements spanning across neurons [143]. Again, this degree of entanglement would be expected to be less than that of tubulin in the same microtubule.

The fourth degree of neural interaction would involve entanglement within highly interconnected cortical areas (e.g., cortical areas of the same sen-

⁴⁸ Calculations that include brain temperature and decoherence issues indicate that quantum coherence is possible in brain (see Hagan et al., [43]) in spite of claims to the contrary [125].

sory modality). Entanglement can occur over long distances when there is a classical channel of information and a pre-existing entanglement. Qubits performing local operations can even send information to entangled qubits via classical channels⁴⁹. This kind of long-range quantum entanglement relies on a physical connection between the brain regions, for example, axonal connectivity between cortical areas or electromagnetic energy flow from one brain region to another. Microtubules binding to nCAMs that traverse the synapse may provide a physical link between the cytoskeleton in sending and receiving neurons.⁵⁰ This degree of entanglement might be expected to be medium to strong because of possible nCAM linkages between microtubules, but also limited to couplings between small groups of subsynaptic zones in widely different brain regions because of the limited nature of point-to-point connections.

The fifth level of neural interaction would involve entanglement among cortical areas having few or no axonal connections. Brain-wide entanglement might occur in a manner predicted by Bell's theorem stating that whenever a quantum measurement is made on one part of a holistic quantum system, this will produce an effect on other parts of the system [7]. This degree of entanglement would be weakest, thereby allowing for the great flexibility in brain response. The overall picture envisioned here is that there are weak, but widespread, couplings between microtubules in unrelated brain regions, with these microtubules, in turn, being strongly coupled to other microtubules in the same and in neighboring neurons and on the basis of specific long-distance connections. The grand total of microtubules involved at any instance may be quite large, but the grand scheme has an essential, self-defining architecture.

Regarding the basis of brain-wide entanglements, the brain arguably *is* a holistic system of directly and indirectly interacting parts, capable of entanglement⁵¹. Many anatomical and physiological factors, other than electrophysiology, contribute to the entire brain behaving as a holistic system. The skull rigidly encases the brain and limits hemodynamics; this in turn affects metabolic processes in an interactive manner. Since tissue mass and fluid volume needs to be kept constant, increased blood flow to one brain region must, of necessity, result in decreased blood flow to another brain region.

The skull protects the brain from some outside forces, albeit imperfectly, leaving neurons sensitive to external magnetic and gravitational fields, both of which can affect the biophysical state of microtubules [98, 102, 127]. While strong magnetic fields are safe for humans, transcranial magnetic stimulation (TMS)

⁴⁹ Although comparisons to neurons have not been made, entangled pairs can effectively communicate by sending 2 bits of information through a classical channel [16]

⁵⁰ Specific nCAMs bind actin, tubulin and actinin [15].

⁵¹ Although entanglements are generally limited to physically interacting systems, it is relevant to note here that entanglement can also be induced between non-interacting particles (e. g., Fattal et al., [28]).

increases neuronal excitability and can sometimes disrupt human cognitive processing [100]. Repetitive TMS, which has a more robust effect than single pulse TMS, has been shown to affect learning [37] and visual cognition [4]. Microgravity has negative effects on cognition and performance [8], but these skills quickly return in a normal gravity environment [126].

Neurons, especially the large pyramidal neurons, also generate electromagnetic activity due to their membrane potentials [70]. Thus, microtubules can, in principle, be affected by electromagnetic activity generated by nearby neurons. The geometry of cortical pyramidal cells may be significant in both the role of sender and receiver. Apical dendrites of large pyramidal cells radiate towards the surface in parallel with those from neighboring pyramidal cells. Basilar dendrites from pyramidal cells form a network parallel to the cortical surface. The convolutions of the cerebral cortex enable the spread of electromagnetic activity to skip over buried areas of cortex. It is not known how far weak electromagnetic activity can spread, but synchronous electromagnetic activity appears capable of having long-distance effects. It is indeed synchronous neuromagnetic activity in distant brain regions that accompanies consciousness [117].

In summary, the human brain defines a volume of space (on average 1100 cm^3) in which many types of interaction are possible. Space has a minimum nonzero volume of approximately 10^{-99} cubic centimeters (one cubic Plank length); this translates into 10^{102} cubic Planck lengths in the human brain. Since quantum states of entanglement in the human brain would be influenced by *all* possible factors – including electrophysiological (membrane potentials, action potentials), mechanical (transport along microtubules), biochemical (phosphorylation of MAP2; detyrosination, acetylation and polyglutamylation of microtubules), biophysical (microtubule polymerization), metabolic (energy utilization), electromagnetic and gravitational – one can argue that it can only be the state of brain-wide quantum coherence that corresponds with the cognitive state. Since tubulin and MAP2 compose a sizable percentage of brain protein, microtubules would be expected to contribute greatly to the equation. The presence of microtubules in the sub-synaptic space and their ability to control synapse parameters only increases their importance.

The notion of brain-wide quantum entanglement depends on quantum entanglements scaling up to mesoscopic and macroscopic ranges. Ghosh et al. [32] provide data that quantum entanglement can scale up for a lithium magnetic salt compound. In a review of the Ghosh paper, Veral [131] notes with great enthusiasm: “a very small amount of entanglement can produce significant effects in the macroscopic world”. If this principle holds in brain, then quantum entanglement may prove highly successful at unraveling some of the greatest mysteries of the human mind.

For one, nonlocal quantum entanglement could explain how biophysical states among widespread microtubules can be unified into memory. Entan-

glement could be responsible for microtubules in different subsynaptic zones undergoing coherent polymerization/depolymerization cycles and for transport of synaptic proteins, such that a select composite of synapses are coherently activated. By controlling inputs, the output of the neuron is also under the control of its coherent microtubule network. If microtubules are particularly predisposed to quantum coherence during higher cognition, then all the information about a synapse's history stored in the subsynaptic microtubules could be shared by quantum coherence with other microtubules storing histories for other synapses.

Quantum entanglements also provide explanations for mental imagery that arises in the absence of sensory inputs, such as in dreams. One of the most widely accepted theory of dreams, the *activation-synthesis hypothesis* of Hobson and McCarley [54], does not fully explain the content of dreams beyond random neuronal firing that triggers the brain to generate stories. Quantum-entanglement theories of the mind get to the issue of how the brain generates those stories, as well as explaining the more enigmatic elements of perception (see Woolf and Hameroff, [143]). The prediction is that dream content is produced by sequences of quantum coherence among microtubules in various subsynaptic zones, linking sets of synapses that have previously shared histories. Furthermore, each component element of the temporal pattern (i. e. the dream plot) would be stored as a separate history, and presumably this would involve quantum entanglement among relevant subsynaptic zones over segments of time.

3.6 Conclusion

This chapter proposes that there is a novel way in which the brain may be unified, apart from traditional synaptic connections. It has long been known that the brain is highly interconnected through axons; nonetheless, these connections are most often confined to a single modality and connect brain areas in a restricted point-to-point fashion. Unified activity among brain microtubules is presently suggested to fill the void. The brain contains many neurotransmitters that affect the neuronal membrane potential; however, it is also true that these neurotransmitters affect the biophysical state of microtubules through ionic fluxes, actions on actin filaments and the phosphorylation of MAP2. Physical responses of microtubules include increased and decreased polymerization, transport and the regulation of receptor proteins inserted into the membrane. Microtubules in the subsynaptic space have properties making them more capable of storing information about synaptic activity than the constituents of the synapses themselves. Microtubules may also play a role in consciousness, thereby enabling attention, perception and dreams. The ubiquitous microtubule arguably sits at the interface between memory and consciousness.

Acknowledgement. I wish to thank my current collaborators: Stuart Hameroff, Jack Tuszynski and Avner Priel for intellectual discussions. Special thanks are due Adele Behar for supporting some of this work and to Malcolm Dean for sending me interesting papers.

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